Assembly: Data Movement Instructions

- CMPE310/CMSC313
- Dr. Kidd

Acknowledgements

- These lectures are duplicates of Dr. Kidd's CMPE310 lectures.
- These lectures are largely from Dr. Chintan Patel, UMBC, with significant modifications and additions from myself.

Intel Assembly

Data Movement Instruction:

- mov (covered already)
- push, pop
- lea (mov and offset)
  - lds, les, lfs, lgs, lss
  - movs, lods, stos
- ins, outs
- xchg, xlat
- lahf, sahf (not covered)
- in, out
- movsx, movzx
- bswap
- cmov

Stack

Stack Instructions

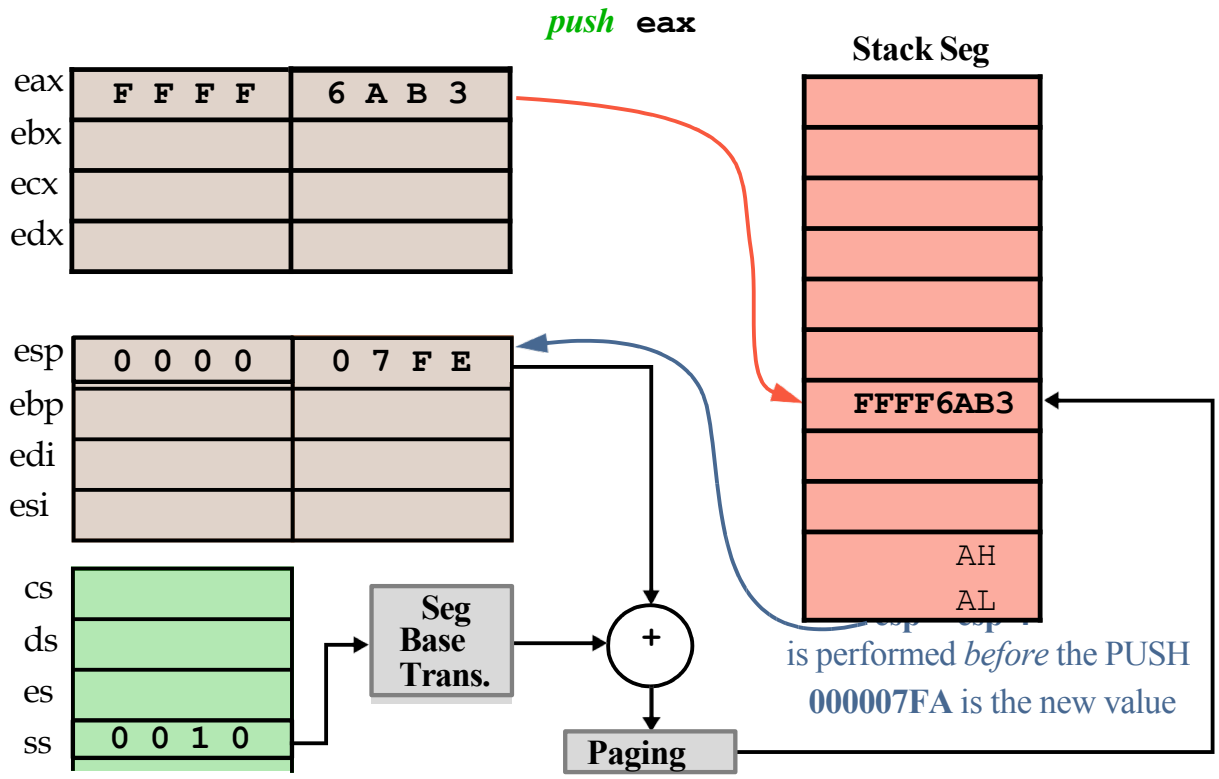
There are six forms of the push and pop instructions.

Register, memory (memory-to-memory copy), immediate, segment register, flags, and all registers

- ***push***:
  - The source of the data may be:
    - Any 16- or 32-bit register, immediate data, any segment register, any word or doubleword of memory data
    - ***pushad*** pushes **eax**, **ecx**, **edx**, **ebx**, **esp**, **ebp**, **edi** and **esi** where the value of **esp** saved on the stack is its value before the ***pushad***.

- ***pop***:
  - The source of the data may be:
  - Any 16- or 32-bit register, any segment register (except for **cs**), any word or doubleword of memory data.

Stack Instructions: push



Address Loading

Address Loading Instructions

Load-Effective Address.

- ***lea***:
  - Loads any 32-bit register with the address of the data, as determined by the instruction addressing mode.
- ***lds and les***:
  - Load a 32-bit offset address and then **ds** or **es** from a 48-bit memory location.
- ***lfs***, ***lgs*** and ***lss*** (80386 and up):
  - Load any 32-bit offset address and then **fs**, **gs** or **ss** from a 48-bit memory location.

	lea	*100(%ebx,%ecx,4),%eax*	#Loads eax with computed address
	lds	*LIST,%edi*	#Loads edi and ds
	lfs	*DATA1,%esi*	#Loads esi and fs

NOTE: ***lea*** calculates the **ADDRESS** given by the right arg and stores it in the left arg!

Address Loading Instructions

Load-Effective Address.

- ***lea*** versus ***mov***:

	lea	*(%edi),%ebx*	#Load the contents of edi into ebx. (1)
	mov	*(%edi),%ebx*	#Load the value at edi into ebx. (2)
	mov	*%edi,%ebx*	#Move the contents of edi into ebx. (3)

- 1 and 3 are equivalent.

- So what are the differences?
  - 3 is faster than 1 and is preferred.
  - However, ***lea*** only works with single args and cannot be used with LIST(%**edi)**.
  - ***mov*** can take any address, e.g., ***mov*** (%**ebx**,%**edi**),%**esi** .

Strings

String Operations

- ***movs*, *lods*, *stos*, *ins*, *outs***
  - Allow data transfers of a byte, a word or a double word, or if repeated, a block of each of these.
  - The *D* flag-bit (direction), **esi** and **edi** are implicitly used.
    - D = 0: Auto increment **edi** and **esi**.
      - Use ***cld*** instruction to clear this flag.
    - D = 1: Auto decrement **edi** and **esi**.
      - Use ***std*** instruction to set it.
  - **edi**:
    - Accesses data in the extra segment. Can NOT override.
  - **esi**:
    - Accesses data in the data segment. Can be overridden with segment override prefix.

String Operations

lods:

- Loads **al**, **ax** or **eax** with data stored at the data segment (or extra segment) + offset
- given by **esi**.
- **esi** is incremented or decremented afterwards:

	lodsb		`#al=ds:(esi)# esi=esi+/-1`
	lodsd		`#eax=ds:(esi)# esi=esi+/-4`
`es`	***lodsb***	*DATA1*	`#Override ds.`

stosb:

- Stores **al**, **ax** or **eax** to the extra segment (es) + offset given by **edi**. es cannot be overridden.
- **edi** is incremented or decremented afterwards:

	stosb		`#es:(edi)=al# edi=edi+/-1`
	stosd		`#es:(edi)=eax# edi=edi+/-4`

String Operations

rep prefix:

- Executes the instruction **ecx** times.

	mov	$0,%edi	#Offset 0.
	mov	$(25*80),%ecx	#Load count.
	mov	*$0x0720,%eax*	#Load value to write
	rep	stosw	

NOTE: ***rep*** does not make sense with the ***lodsb*** instruction.

movs:

- Moves a byte, word or doubleword from data segment and offset **esi** to extra segment and offset **edi**.
- Increments/decrements both **edi** and **esi**:

	movsb		#es:(edi)=ds:(esi)# edi+/-=1# esi+/-=1
	movsd		#es:(edi)=ds:(esi)# edi+/-=4# esi+/-=4

String Operations and Exchange

- ***ins/outs***:
  - Transfers a byte, word or doubleword of data from/to an I/O device into/out of the
  - extra/data segment + offset **edi/esi**, respectively. The I/O address is stored in the **edx** register.

	insb		`#es: (edi)=(edx)# edi+/-=1`
	insd		`#es: (edi)=(edx)# edi+/-=4`
	insw		`#es: (edi)=(edx)# edi+/—=2# esi+/==2`
	outsb		`#(edx)=ds:(esi)# esi=esi+/-1`

- ***xchg***:
  - Exchanges the contents of a register with the contents of any other register or memory location.
  - It can NOT exchange segment registers or memory-to-memory data.
  - Byte, word and doublewords can be exchanged using any addressing mode (except immediate, of course).

	xchg	`%esi,%edx`	`#Exchange edx and esi`

Miscellaneous

Miscellaneous Data Transfer Operations

- ***movsx*** and ***movzx*** (80386 and up only):
  - Move-and-sign-extend and Move-and-zero-extend:

	movsx	%bl,%cx	#Sign-extends bl into cx
	movzx	DATA2,%eax	#Zero extends word at DATA2 in eax.

- ***bswap*** (80486 and up only):
  - Swaps the first byte with the forth, and the second byte with the third.
  - Used to convert between little endian and big endian:

- ***cmov*** (Pentium and up only):
  - These instructions move data only if a condition is true.
  - Conditions are set by a previous instruction and include *Carry*, *Zero*, *Sign*, *Over- flow* and *Parity*:

	cmovz	%ebx,%eax	#Move if Zero flag is set else do nothing

  - There are many variations of this instruction (see intel instructions doc or text).

Assembler Directives

- **Segment Override Prefix:**
  - Allows the programmer to override the default segment.

	mov	(0x1234),%ax	#moves from DS:0x1234 (default)
	mov	es:(0x1234),%ax	#moves from ES:0x1234 (override)

- **Procedure Calls**

	push	args...	#Push the arguments on stack.
	call	procedure_name	#Call the procedure
	add	*esp,#*	#Restore the stack pointer.